

CARL-WSN: Context-Aware Reinforcement Learning for Hybrid Routing in Heterogeneous IoT-Driven Wireless Sensor Networks

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Abstract

Wireless Sensor Networks (WSNs) deployed in IoT-driven smart city environments carry heterogeneous application traffic with fundamentally different quality-of-service requirements. Existing routing protocols — including recent reinforcement learning (RL) based approaches — optimise a single objective such as energy efficiency or network lifetime without adapting routing behaviour to the urgency class of individual packets. This paper proposes CARL-WSN, a Context-Aware Reinforcement Learning protocol that uses a tabular Q-Learning agent to dynamically select the routing paradigm — proactive, reactive, or hybrid — on a per-packet basis. The agent observes a three-dimensional context vector comprising application urgency class, normalised residual energy, and link stability score, and learns a class-dependent multi-objective reward function that weights latency, overhead, and energy preservation according to traffic priority. A safety constraint forces Class A emergency packets onto proactive routing regardless of the agent’s policy, guaranteeing sub-100 ms delivery at all times. CARL-WSN further integrates energy-aware relay selection and bounded critical-node packet deferral to extend network lifetime. The compact Q-table (216 bytes of RAM, stored in single precision) enables deployment on resource-constrained sensor hardware. CARL-WSN is evaluated in MATLAB against seven baseline protocols —

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AODV, DSDV, ZRP, two domain-specific baselines (EH-Routing, MSLBA), and two recent RL-based approaches (RLCR and FQ-UCR) — under four traffic scenarios across 20 independent seeds (640 runs), with all protocols re-implemented on a common distance-dependent lossy-channel model so that delivery, latency, and energy emerge from identical physical assumptions. Under mixed traffic, CARL-WSN extends network lifetime beyond all baselines (first node death at round 418, the highest of all protocols; half-node death at round 2198, second only to MSLBA), delivers the lowest Class A emergency latency among the reactive and learning-based protocols (24.2 ms, a 25.6% reduction versus AODV), and incurs routing overhead far below the flooding- and table-based protocols (0.53 control packets per data packet, versus 14.8 for AODV and 2.31 for DSDV). These gains are obtained by trading a modest amount of packet delivery: CARL-WSN achieves 90.2% PDR under its tuned deferral policy, within roughly five points of the baselines, with energy balance and throughput statistically comparable across all protocols. The learned Q-table converges within approximately 1150 rounds, and the agent autonomously discovers context-dependent policies that adapt routing overhead to traffic-class composition — a capability absent in all seven baselines. A sensitivity analysis confirms that CARL-WSN’s performance is robust to its principal design parameters.

Keywords: Wireless sensor networks, Reinforcement learning, Q-Learning, Context-aware routing, Hybrid routing protocol, IoT, Smart city, Energy-aware relay selection, Heterogeneous traffic

1. Introduction

The proliferation of Internet of Things (IoT) devices in smart city infrastructure has transformed Wireless Sensor Networks (WSNs) from single-application monitoring systems into multi-service platforms that simultaneously carry traffic with fundamentally different quality-of-service (QoS) requirements [1, 2]. A single WSN deployment may generate emergency alerts requiring sub-100 ms delivery, periodic telemetry tolerating delays of tens of seconds, and event-driven uploads with best-effort requirements. This traffic heterogeneity demands routing protocols that adapt their behaviour to the urgency class of individual packets — a capability absent from both classical and recent learning-based routing approaches.

Routing protocols for WSNs fall into three paradigms, each with inherent

trade-offs. Proactive protocols such as DSDV [3] maintain routing tables continuously, delivering low latency at the cost of high energy overhead from periodic broadcasts. Reactive protocols such as AODV [4] and DSR [5] discover routes on demand, conserving energy in low-traffic conditions but introducing route-discovery delays of 20–100 ms that violate emergency latency constraints [6]. Hybrid protocols such as ZRP [7] partition the network into zones and apply different modes within and across boundaries, but use a fixed zone radius that does not adapt to node energy state or traffic urgency.

Recent work has applied reinforcement learning (RL) to WSN routing with promising results. Daanounne and Baghdad [8] proposed RLCR, a Q-Learning-based clustering and routing protocol that optimises cluster head selection and inter-cluster forwarding for energy efficiency. Wang and Duan [9] proposed FQ-UCR, combining fuzzy logic for unequal cluster formation with Q-Learning for inter-cluster routing. While both protocols demonstrate the potential of RL for adaptive WSN routing, they share a fundamental limitation: routing decisions are made based on network-level metrics (residual energy, distance to base station) without any awareness of the application-level urgency class of the packet being routed. An emergency alert and a routine temperature reading receive identical routing treatment — a systematic inefficiency that leads to either wasted energy on low-urgency traffic or dangerous delays on safety-critical packets.

A 2025 comprehensive survey by Khedr et al. [10] explicitly identified dynamic application-context-based hybrid mode switching as an open research direction in IoT-driven WSNs, noting that reactive and hybrid context-aware approaches have received significantly less attention than proactive ones. No published protocol uses reinforcement learning to select between fundamentally different routing paradigms — proactive, reactive, and hybrid — on a per-packet basis according to application urgency. This paper addresses that gap directly.

The main contributions of this paper are as follows:

- A formal Markov Decision Process (MDP) formulation for routing paradigm selection in heterogeneous IoT-WSNs, with an $18\text{-state} \times 3\text{-action}$ state-action space derived from a three-dimensional context vector comprising application urgency class U , normalised residual energy E_r , and link stability score L_s .
- CARL-WSN, a Q-Learning-based routing protocol that learns the optimal mapping from context states to routing paradigms (proactive,

reactive, or hybrid) through online interaction with the network environment, using a class-dependent multi-objective reward function that weights latency, overhead, and energy preservation according to traffic priority. A safety constraint forces Class A emergency packets to proactive routing, guaranteeing sub-100 ms delivery regardless of the agent’s learned policy.

- An energy-aware relay selection mechanism combined with critical-node packet deferral that extends network lifetime by preferring high-residual-energy next-hop nodes and deferring non-urgent packets when the transmitting node is near energy depletion — a context-aware lifetime extension capability that only urgency-aware protocols can implement.
- A comprehensive evaluation against seven baselines — three classical protocols (AODV, DSDV, ZRP), two domain-specific baselines (EH-Routing, MSLBA), and two recent RL-based protocols (RLCR, FQ-UCR) — across four traffic scenarios and 20 independent seeds (640 runs). Crucially, all eight protocols are re-implemented on a common distance-dependent lossy-channel model with per-hop acknowledgement and retransmission, so that packet delivery, latency, and energy consumption emerge from identical physical assumptions rather than from protocol-specific idealisations. Under this unified methodology, CARL-WSN extends network lifetime beyond all baselines (highest first-node-death round) and attains the lowest emergency-traffic latency among the reactive and learning-based protocols, while incurring routing overhead far below the flooding- and table-based protocols. These gains are obtained by trading a modest, quantified reduction in packet delivery, which we characterise explicitly.

The remainder of this paper is organised as follows. Section 2 reviews related work on hybrid, context-aware, and RL-based routing in WSNs. Section 3 presents the system model. Section 4 describes the CARL-WSN protocol design, including the MDP formulation and Q-Learning mechanism. Section 5 presents the simulation setup. Section 6 presents and analyses the results. Section 7 discusses key findings, trade-offs, and limitations. Section 8 concludes the paper.

2. Related Work

2.1. Proactive and Reactive Routing Protocols

Routing protocols for WSNs have been extensively surveyed and classified [11]. Proactive protocols maintain routing tables continuously through periodic broadcasts. DSDV [3] established the foundational destination-sequenced distance-vector approach, guaranteeing loop-free routes at all times. OLSR [12] extended this paradigm through multipoint relay selection, substantially reducing flooding overhead in dense networks. While proactive protocols deliver consistently low latency by eliminating route discovery delays, their continuous maintenance overhead leads to significant energy waste on low-traffic nodes [11].

Reactive protocols discover routes exclusively on demand. AODV [4] pioneered this approach by initiating route requests only when data must be transmitted, conserving energy in sparse or low-traffic conditions. However, the route discovery delay of 20–100 ms introduced by RREQ/RREP flooding renders reactive protocols unsuitable for latency-sensitive applications [6]. A comparative analysis by Nurul Hidayah et al. [13] confirmed that neither proactive nor reactive protocols alone achieve optimal performance across varying network conditions, motivating the need for adaptive hybrid approaches. Recent work [14] demonstrated that AODV, DSDV, and DSR all exhibit significant performance degradation under heterogeneous traffic loads, highlighting the fundamental limitation of single-mode routing for IoT-driven smart city deployments.

2.2. Hybrid Routing Protocols

Hybrid protocols attempt to combine the latency advantages of proactive routing with the energy efficiency of reactive routing. ZRP [7] pioneered zone-based hybrid routing by maintaining proactive routes within a fixed-radius zone and using reactive inter-zone routing for destinations beyond the zone boundary. While ZRP reduces control overhead compared to purely proactive protocols, its fixed zone radius is configured statically and does not adapt to node energy state or traffic urgency.

Subsequent hybrid approaches incorporated additional optimisation dimensions. Chaurasiya et al. [15] proposed a hybrid clustering technique for multilevel heterogeneous IoT WSNs, combining cluster-based organisation with multi-hop routing to reduce energy consumption. Lei [16] proposed a hybrid energy-aware approach combining particle swarm optimisation with fuzzy

119 clustering for IoT routing, demonstrating 52% energy reduction compared
 120 to DEEC, though at the cost of requiring offline training data. Nadanam
 121 et al. [17] recently proposed a quantum-optimised hybrid routing framework
 122 achieving 30% energy reduction, though the computational complexity of
 123 quantum optimisation makes it unsuitable for resource-constrained sensor
 124 hardware.

125 A common limitation across all these hybrid approaches is that mode
 126 selection or routing decisions are made based on a single optimisation objective
 127 — typically energy or load — without awareness of the application-level urgency
 128 of the traffic being routed.

129 *2.3. Context-Aware Routing in IoT-WSNs*

130 Context-aware computing for IoT was formally defined by Perera et al. [18],
 131 who identified five contextual dimensions — identity, location, activity, time,
 132 and relations — as fundamental to intelligent IoT decision making. This
 133 foundational work motivates the use of contextual information in routing
 134 decisions, though the specific application to RL-based WSN hybrid routing
 135 mode selection had not been addressed until the present work.

136 Context-aware routing has since been applied across several IoT and WSN
 137 domains. Monteiro et al. [19] proposed context-aware network selection in
 138 heterogeneous wireless networks, demonstrating that context-driven decisions
 139 outperform static policies, but their work focused on network selection rather
 140 than intra-WSN routing mode switching. Sithik and Kumar [20] proposed
 141 a multi-context aware parent selection scheme for the RPL protocol using
 142 grey set theory, incorporating multiple contextual parameters into routing
 143 metric computation. However, their approach targets RPL parent selection
 144 exclusively and does not implement dynamic switching between fundamentally
 145 different routing paradigms based on application urgency class. Karagiannis
 146 et al. [21] proposed context-aware routing for fog computing systems, reducing
 147 latency by 23% and hop count by 73%, though their work operates at the fog
 148 layer rather than the WSN sensor layer and does not model heterogeneous
 149 traffic classes with conflicting QoS requirements. Hasan et al. [22] proposed
 150 a context-aware trust prediction framework for opportunistic IoT routing
 151 using Bayesian inference, achieving 88.9% delivery ratio, though the trust-
 152 based context dimension differs fundamentally from the urgency-energy-
 153 stability context vector proposed in CARL-WSN. Akpakwu et al. [23] proposed
 154 a context-aware congestion control scheme for CoAP/UDP IoT networks,
 155 applying context sensing to transport-layer decisions rather than routing.

Smart city IoT routing has also received significant attention. Karunkuzhali et al. [24, 25] proposed QoS-aware routing frameworks for IoT-WSNs in smart cities, demonstrating that multi-class service differentiation improves network performance. However, none of these works implement dynamic routing paradigm switching based on per-packet application urgency. El-Fouly et al. [26] proposed environment-aware real-time routing for multi-sink WSNs in smart cities, incorporating deadline constraints and link quality metrics. While this work shares the multi-sink topology with our prior work [27] and considers real-time delivery requirements. Smart city IoT routing has also been addressed by Hanif et al. [28] through opportunistic IoT-WSN integration, Inga et al. [29] through WSN sizing and routing optimisation, and Ekler et al. [30] through energy-aware IoT routing algorithms. However, none of these works implement dynamic routing paradigm switching based on per-packet application urgency. It does not model heterogeneous application urgency classes or implement context-driven mode switching.

2.4. Reinforcement Learning for WSN Routing

The application of reinforcement learning to WSN routing has emerged as one of the most active research frontiers in recent years [31, 32, 33]. RL-based protocols offer the ability to learn routing policies through online interaction with the network environment, adapting to changing conditions without requiring explicit models or predefined rules.

Daanoune and Baghdad [8] proposed RLCR, a reinforcement learning-based clustering and routing protocol for heterogeneous WSNs. RLCR uses Q-Learning in two phases: first for cluster head (CH) selection based on residual energy, distance to the base station, and node density; and second for inter-cluster multi-hop routing where each CH learns the optimal next-hop CH toward the base station. RLCR demonstrated significant lifetime improvements over LEACH and DEEC variants. However, RLCR does not differentiate traffic by application urgency class — all packets receive identical routing treatment regardless of whether they carry emergency alerts or routine telemetry. Furthermore, RLCR’s RL agent operates exclusively within the clustering paradigm and does not consider switching between fundamentally different routing modes.

Wang and Duan [9] proposed FQ-UCR, a hybrid approach combining fuzzy logic (FL) for unequal cluster head selection with Q-Learning for inter-cluster routing. FQ-UCR addresses the hotspot problem by forming smaller clusters near the base station and larger clusters at greater distances, using a fuzzy

Table 1: Comparison of CARL-WSN with Related Routing Works

Protocol	AI/ML method	Context dims	Mode switching	Traffic classes	Urgency aware	Energy-aware relay	Paradigm switching
RLCR [8]	Q-Learning	3	CH selection	1	No	No	No
FQ-UCR [9]	Fuzzy + Q-Learning	3	CH selection	1	No	No	No
Sithik & Kumar [20]	Grey set theory	2	RPL parent	1	No	No	No
Karagiannis et al. [21]	None	2	Fog forwarding	1	No	No	No
Hasan et al. [22]	Bayesian	1 (trust)	Route selection	1	No	No	No
ZRP [7]	None	1 (zone)	Hybrid fixed	1	No	No	No
CARL-WSN (Ours)	Q-Learning	3	Per-packet RL	3	Yes	Yes	Yes

inference system to compute CH selection probability. The Q-Learning agent then selects the optimal forwarding CH for inter-cluster data relay. FQ-UCR demonstrated improved energy efficiency and lifetime compared to EEUC and CHEF. Similar to RLCR, however, FQ-UCR does not model heterogeneous traffic classes or adapt routing behaviour to application urgency.

Sudha et al. [34] combined Q-Learning with machine learning for cluster head selection and routing in 5G-enabled WSNs, demonstrating improved throughput over LEACH, but without considering per-packet application urgency as a routing criterion. Saadallah and Alabady [35] identified energy-efficient context-sensitive routing as an open challenge in their comprehensive IoT routing study, while Kumar et al. [36] confirmed through bibliometric analysis that application-context awareness remains an emerging research gap in energy-efficient IoT-WSN routing.

Table 1 summarises the key distinguishing features of CARL-WSN compared to the most closely related works. CARL-WSN is the only protocol that combines RL-based learning with per-packet application urgency awareness for full routing paradigm switching.

2.5. Research Gaps

The preceding review reveals four specific gaps that motivate CARL-WSN. First, Khedr et al. [10] conducted the first comprehensive survey of context-aware routing in IoT-driven WSNs and explicitly identified dynamic application-context-based mode switching as an open research direction, noting that reactive and hybrid context-aware methods have received significantly less attention than proactive approaches. Second, existing RL-based WSN routing protocols (RLCR [8], FQ-UCR [9]) apply Q-Learning within a single routing paradigm (clustering) without awareness of application urgency class — the RL agent cannot distinguish between emergency and routine traffic. Third,

220 Chu et al. [37], Khanna et al. [38], and Prashanth et al. [39] all noted that routing
 221 adaptation in heterogeneous WSNs remains dominated by single-objective
 222 optimisation approaches that cannot simultaneously satisfy the conflicting
 223 requirements of heterogeneous application traffic classes. Fourth, no published
 224 protocol uses RL to learn which routing paradigm (proactive, reactive, or hybrid)
 225 to use for each packet based on a joint context vector of urgency, energy,
 226 and link stability. Fifth, no RL-based WSN routing protocol incorporates a
 227 safety constraint that guarantees latency bounds for critical traffic regardless
 228 of the learned policy — a requirement for safety-critical IoT deployments.

229 3. System Model

230 3.1. Network Topology

231 Consider a WSN comprising N heterogeneous sensor nodes deployed
 232 randomly over a square area of side length A m according to a uniform
 233 distribution. Let $\mathcal{N} = \{1, 2, \dots, N\}$ denote the set of all nodes. A base
 234 station (BS) is located at the centre of the deployment area, and a secondary
 235 sink node is positioned at the right boundary, consistent with the multi-
 236 sink topology in our prior work [27]. Both the BS and sink are assumed
 237 to have unconstrained energy. Each sensor node $i \in \mathcal{N}$ is characterised by
 238 a fixed position (p_i^x, p_i^y) and a heterogeneous initial energy E_0^i drawn from
 239 a uniform distribution $E_0^i \sim \mathcal{U}(E_{\min}, E_{\max})$, reflecting the reality of diverse
 240 sensor hardware in smart city deployments [40].

241 A node i is considered alive if its residual energy $E_i(t) > 0$ at simulation
 242 round t , and dead otherwise. Dead nodes neither transmit nor receive. The
 243 network is considered operational as long as at least 10% of nodes remain
 244 alive.

245 3.2. Traffic Model

246 Smart city IoT deployments generate heterogeneous application traffic
 247 with fundamentally different QoS requirements. This paper models three
 248 traffic classes simultaneously:

- 249 • **Class A — Emergency traffic:** Generated by safety-critical sensors
 250 such as fire detectors, intrusion alarms, and accident monitors. Modelled
 251 as a Poisson process with arrival rate λ_A . Strict latency requirement:
 252 end-to-end delay must not exceed 100 ms.

- 253 • **Class B — Periodic telemetry:** Generated by environmental sensors
 254 reporting temperature, humidity, air quality, and energy consumption.
 255 Modelled as a deterministic periodic process with one packet per minute.
 256 Tolerates delays up to 30 seconds.
- 257 • **Class C — Event-driven uploads:** Generated by motion detectors,
 258 occupancy sensors, and camera triggers. Modelled as a Poisson process
 259 with arrival rate λ_C . Best-effort delivery with no hard latency constraint.

260 Each packet carries a class tag $U \in \{A, B, C\}$ in its header, which is read
 261 by the CARL-WSN context classifier at the transmitting node before routing
 262 decisions are made.

263 3.3. Channel and Energy Model

264 Wireless communication is modelled using the first-order radio energy
 265 model [41], which captures the dominant sources of energy consumption in
 266 sensor nodes: electronics and amplification. The energy consumed by node i
 267 to transmit an L -bit packet to a receiver at distance d is:

$$E_{tx}(L, d) = \begin{cases} L \cdot E_{elec} + L \cdot \varepsilon_{fs} \cdot d^2 & d < d_0 \\ L \cdot E_{elec} + L \cdot \varepsilon_{mp} \cdot d^4 & d \geq d_0 \end{cases} \quad (1)$$

268 where $d_0 = \sqrt{\varepsilon_{fs}/\varepsilon_{mp}} = 87.71$ m is the crossover distance between free-
 269 space and multipath propagation models. The energy consumed to receive
 270 an L -bit packet is:

$$E_{rx}(L) = L \cdot E_{elec} \quad (2)$$

271 The residual energy of node i after a transmission round is:

$$E_i(t+1) = E_i(t) - E_{tx}(L, d) - E_{rx}(L) \quad (3)$$

272 subject to $E_i(t) \geq 0$.

273 3.4. Link Stability Model

274 To capture dynamic link quality variations, CARL-WSN maintains a per-
 275 node link stability score $L_s^i(t) \in [0, 1]$, computed as an Exponential Weighted
 276 Moving Average (EWMA) of the ACK success history over a sliding window
 277 of W most recent transmissions:

$$L_s^i(t) = \sum_{k=0}^{W-1} w_k \cdot a_k^i \quad (4)$$

where $a_k^i \in \{0, 1\}$ is the ACK outcome of the k -th most recent transmission from node i , and $w_k = \alpha(1 - \alpha)^k / Z$ are normalised exponentially decaying weights with smoothing factor $\alpha = 0.3$ and normalisation constant $Z = \sum_{k=0}^{W-1} (1 - \alpha)^k$.

3.5. Energy-Aware Relay Selection Model

CARL-WSN introduces an energy-aware relay selection mechanism that extends network lifetime by preferring next-hop nodes with higher residual energy. When forwarding a packet, node i selects the next-hop relay j^* from its set of alive neighbours $\mathcal{N}_i = \{j : d_{ij} \leq R_{tx}, E_j(t) > 0\}$ by maximising a composite score:

$$j^* = \arg \max_{j \in \mathcal{N}_i} \left[w_p \cdot \frac{d_{iBS} - d_{jBS}}{d_{iBS}} + w_e \cdot \frac{E_j(t)}{E_0^j} \right] \quad (5)$$

where d_{iBS} and d_{jBS} are the distances from nodes i and j to the base station respectively, $w_p = 0.4$ is the progress weight, and $w_e = 0.6$ is the energy weight. The first term encourages geographic progress toward the BS, while the second term favours relays with higher residual energy ratios, distributing the forwarding load away from energy-depleted nodes. Only candidates with positive progress ($d_{jBS} < d_{iBS}$) are considered, ensuring packets always advance toward the BS.

3.6. Critical-Node Packet Deferral

When a node's residual energy ratio falls below a critical threshold $E_r^i < T_{defer} = 0.15$, CARL-WSN defers transmission of non-urgent packets (Class B and Class C) to preserve the node's remaining energy for potential emergency traffic. Deferred packets incur only a minimal listening energy cost of $0.1 \cdot L \cdot E_{elec}$. Class A emergency packets are never deferred regardless of node energy state — the safety constraint takes absolute priority. This deferral mechanism is a unique capability of urgency-aware protocols: baseline protocols that do not distinguish traffic classes cannot selectively defer low-priority traffic to extend node lifetime.

305 3.7. Performance Metrics

306 Eight metrics are used to evaluate protocol performance:

307 **Network lifetime:** The First Node Death round $r_{FND} = \min\{t :$
 308 $\exists i, E_i(t) = 0\}$ and the Half Node Death round $r_{HND} = \min\{t : |\{i : E_i(t) =$
 309 $0\}| \geq N/2\}$.

310 **Packet Delivery Ratio:**

$$PDR = \frac{D_{delivered}}{D_{total}} \quad (6)$$

311 **Energy balance** (Gini coefficient):

$$G = \frac{2 \sum_{i=1}^N i \cdot c_i}{N \sum_{i=1}^N c_i} - \frac{N+1}{N} \quad (7)$$

312 where c_i is the energy consumed by node i , sorted in ascending order
 313 ($c_1 \leq c_2 \leq \dots \leq c_N$). A lower G indicates more uniform energy consumption.

314 **Routing overhead:**

$$\Omega = \frac{C_{ctrl}}{D_{total}} \quad (8)$$

315 where C_{ctrl} is the total number of routing control packets generated during
 316 the simulation.

317 **Additional metrics:** Mean Class A latency, average energy consumed
 318 per round, and network throughput (packets delivered per round).

319 4. CARL-WSN Protocol Design

320 4.1. Design Motivation

321 Existing RL-based WSN routing protocols apply Q-Learning within a
 322 single routing paradigm. RLCR [8] uses Q-Learning for cluster head selection
 323 and inter-cluster forwarding — both decisions operate within the clustering
 324 paradigm. FQ-UCR [9] similarly applies Q-Learning only to select the next
 325 forwarding cluster head. In both protocols, the fundamental routing mecha-
 326 nism (clustering with multi-hop inter-cluster relay) never changes regardless
 327 of traffic urgency.

328 CARL-WSN addresses a categorically different problem: *which routing*
 329 *paradigm should be used for each packet?* The RL agent learns to select
 330 between proactive, reactive, and hybrid routing — three fundamentally differ-
 331 ent mechanisms with distinct latency, overhead, and energy characteristics —

332 based on a per-packet context vector that includes application urgency class.
 333 This is routing paradigm switching, not routing optimisation within a fixed
 334 paradigm.

335 4.2. The Context Vector

336 For each packet to be transmitted by node i of class U , the context vector
 337 $\mathbf{C}_i = (U, E_r^i, L_s^i)$ is computed, where:

- 338 • $U \in \{A, B, C\}$ is the urgency class read directly from the packet header.
 339 Zero additional computation is required.
- 340 • $E_r^i = E_i(t)/E_0^i \in [0, 1]$ is the normalised residual energy ratio. $E_r^i = 1$
 341 indicates full energy; $E_r^i \rightarrow 0$ indicates near-depletion.
- 342 • $L_s^i \in [0, 1]$ is the link stability score computed using (4).

343 4.3. MDP Formulation

344 The routing paradigm selection problem is formulated as a Markov Decision
 345 Process $(\mathcal{S}, \mathcal{A}, R, \gamma)$ where each packet transmission constitutes one decision
 346 step.

347 4.3.1. State Space

348 The continuous context vector is discretised into a finite state space
 349 suitable for tabular Q-Learning:

- 350 • Urgency class U : 3 values — A (emergency), B (telemetry), C (event-
 351 driven).
- 352 • Residual energy level: 3 levels — low ($E_r^i < 0.3$), mid ($0.3 \leq E_r^i < 0.7$),
 353 high ($E_r^i \geq 0.7$).
- 354 • Link stability level: 2 levels — unstable ($L_s^i < 0.7$), stable ($L_s^i \geq 0.7$).

355 The total state space is $|\mathcal{S}| = 3 \times 3 \times 2 = 18$ states. Each state is encoded
 356 as a linear index:

$$s = (U_{num} - 1) \times 6 + (E_{level} - 1) \times 2 + L_{level} \quad (9)$$

357 where $U_{num} \in \{1, 2, 3\}$, $E_{level} \in \{1, 2, 3\}$, and $L_{level} \in \{1, 2\}$. Table 2
 358 enumerates all 18 states with their physical interpretations.

Table 2: CARL-WSN State Space: All 18 States

State	Real-world meaning	U	E_r level	L_s level
1–2	Emergency, low energy, unstable/stable link	A	low	unstable/stable
3–4	Emergency, moderate energy, unstable/stable	A	mid	unstable/stable
5–6	Emergency, healthy node, unstable/stable	A	high	unstable/stable
7–8	Telemetry, low energy, unstable/stable	B	low	unstable/stable
9–10	Telemetry, moderate energy, unstable/stable	B	mid	unstable/stable
11–12	Telemetry, healthy node, unstable/stable	B	high	unstable/stable
13–14	Best-effort, low energy, unstable/stable	C	low	unstable/stable
15–16	Best-effort, moderate energy, unstable/stable	C	mid	unstable/stable
17–18	Best-effort, healthy node, unstable/stable	C	high	unstable/stable

359 4.3.2. Action Space

360 The agent selects one of three routing paradigms per packet:

$$\mathcal{A} = \{\text{proactive, reactive, hybrid}\} \quad (10)$$

361 The Q-table has dimensions $18 \times 3 = 54$ entries, requiring only $54 \times 4 = 216$
 362 bytes of storage in single-precision floating point — deployable on any sensor
 363 platform with at least 1 KB of RAM.

364 4.3.3. Reward Function

365 After selecting action a in state s , the agent observes the routing outcome
 366 and receives a class-dependent multi-objective reward:

$$R(s, a) = w_1^U \cdot R_{lat} + w_2^U \cdot R_{oh} + w_3^U \cdot R_{en} \quad (11)$$

367 where the three reward components are:

$$R_{lat} = \begin{cases} -1 & \text{if } U = A \text{ and } \ell > 100 \text{ ms} \\ 1 - \ell / \ell_{\max}^U & \text{otherwise} \end{cases} \quad (12)$$

$$R_{oh} = 1 - \min\left(\frac{c_{ctrl}}{n_{nb} \cdot 0.20}, 1\right) \quad (13)$$

$$R_{en} = \min\left(\frac{E_r^{after}}{E_r^{before}}, 1\right) \quad (14)$$

368 Here ℓ is the observed end-to-end latency, $\ell_{\max}^U$ is the class-specific latency
 369 tolerance (100 ms for Class A, 30 s for Class B, ∞ for Class C), c_{ctrl} is the

370 number of control packets generated, n_{nb} is the neighbour count, and E_r^{before}
 371 and E_r^{after} are the residual energy ratios before and after transmission.
 372 The class-dependent weight vectors are:

$$(w_1^U, w_2^U, w_3^U) = \begin{cases} (0.7, 0.15, 0.15) & U = A \text{ (latency dominates)} \\ (0.2, 0.3, 0.5) & U = B \text{ (energy dominates)} \\ (0.1, 0.3, 0.6) & U = C \text{ (energy dominates)} \end{cases} \quad (15)$$

373 This formulation ensures that the RL agent learns different routing pref-
 374 erences for different traffic classes: Class A packets are penalised heavily for
 375 latency violations, while Class B and Class C packets are rewarded primarily
 376 for energy preservation. The weight vectors encode the fundamental QoS
 377 distinction between emergency and non-emergency IoT traffic.

378 4.3.4. Q-Learning Update Rule

379 The Q-table is updated after each packet transmission using the standard
 380 Q-Learning rule:

$$Q(s, a) \leftarrow Q(s, a) + \alpha \left[R(s, a) + \gamma \max_{a'} Q(s', a') - Q(s, a) \right] \quad (16)$$

381 where $\alpha = 0.1$ is the learning rate, $\gamma = 0.9$ is the discount factor, s' is the
 382 next state observed after packet delivery, and $\max_{a'} Q(s', a')$ is the maximum
 383 expected future reward from the next state.

384 Action selection follows an ε -greedy policy:

$$a = \begin{cases} \text{random action from } \mathcal{A} & \text{with probability } \varepsilon \\ \arg \max_a Q(s, a) & \text{with probability } 1 - \varepsilon \end{cases} \quad (17)$$

385 The exploration rate decays exponentially:

$$\varepsilon_{t+1} = \max(\varepsilon_t \cdot \delta, \varepsilon_{\min}) \quad (18)$$

386 with initial $\varepsilon_0 = 0.1$, decay factor $\delta = 0.998$, and minimum $\varepsilon_{\min} = 0.01$.
 387 This schedule ensures sufficient exploration in early rounds while converging
 388 to a near-greedy exploitation policy within approximately 500 rounds.

389 4.4. Safety Constraint

390 A critical design requirement for safety-critical IoT deployments is that
391 emergency traffic must *never* be delayed by route discovery, regardless of
392 what the RL agent has learned. CARL-WSN enforces this through a hard
393 constraint that overrides the agent’s policy:

$$\text{If } U = A : \quad a = \text{proactive} \quad (\text{unconditional}) \quad (19)$$

394 This constraint is applied *before* the ε -greedy selection in (17), ensuring
395 that Class A packets are always routed via proactive mode with pre-built
396 routing tables, guaranteeing sub-100 ms delivery at all times. The Q-table
397 entries for states 1–6 (all Class A states) are never updated by the agent —
398 proactive routing for emergencies is a design invariant, not a learned behaviour.

399 This separation between learned policy (Class B and C) and hardcoded
400 safety guarantee (Class A) is a distinguishing feature of CARL-WSN. Neither
401 RLCR [8] nor FQ-UCR [9] implements traffic-class-specific safety constraints,
402 because neither protocol differentiates traffic by urgency class.

403 4.5. Routing Modes

404 The three routing paradigms available to the RL agent are:

405 4.5.1. Proactive Mode

406 Inspired by DSDV [3], proactive mode maintains routing tables through
407 periodic broadcasts every round. When a packet requires transmission, the
408 next hop toward the BS is selected using the energy-aware relay selection
409 mechanism defined in (5), combining geographic progress with residual energy
410 preference. The energy overhead includes both data transmission cost and
411 a table maintenance overhead of 10% of the data packet energy per round.
412 This mode is mandatory for Class A emergency traffic and is also selected by
413 the RL agent when link instability makes reactive route discovery unreliable.

414 4.5.2. Reactive Mode

415 Inspired by AODV [4], reactive mode initiates route discovery only when
416 a packet needs to be sent. A Route Request (RREQ) is flooded to neighbours
417 within transmission range, and a Route Reply (RREP) is returned along the
418 reverse path with energy-aware relay selection applied at each hop. This
419 eliminates periodic broadcast overhead but introduces route discovery delay
420 of 85–115 ms. The RL agent learns to select reactive mode primarily for

421 Class B and Class C traffic on healthy nodes with stable links, where the
 422 energy saving from avoiding table maintenance outweighs the latency cost.

423 4.5.3. Hybrid Mode

424 Extending ZRP [7], hybrid mode maintains proactive routes within a
 425 dynamically-sized zone and uses reactive inter-zone routing beyond the zone
 426 boundary. The proactive zone radius adapts based on the node's residual
 427 energy ratio:

$$r_{zone}^i = \begin{cases} 120 \text{ m} & E_r^i \geq 0.6 \text{ (healthy)} \\ 80 \text{ m} & 0.3 \leq E_r^i < 0.6 \text{ (moderate)} \\ 40 \text{ m} & E_r^i < 0.3 \text{ (critical)} \end{cases} \quad (20)$$

428 The RL agent learns to select hybrid mode primarily for energy-critical
 429 nodes carrying non-emergency traffic, where the reduced zone radius limits
 430 broadcast overhead while maintaining route availability within the immediate
 431 neighbourhood.

432 4.6. Complete Per-Packet Routing Procedure

433 Algorithm 1 presents the complete per-packet routing procedure, inte-
 434 grating the context vector computation, safety constraint, ε -greedy action
 435 selection, energy-aware routing, packet deferral, and Q-table update.

436 4.7. Overhead Analysis

437 The per-packet control overhead ω of CARL-WSN adapts to the selected
 438 mode:

$$\omega = \begin{cases} n_{nb} \cdot 0.08 & \text{proactive (Class A)} \\ n_{nb} \cdot 0.05 & \text{reactive (Class B/C)} \\ n_{nb} \cdot 0.06 & \text{hybrid} \end{cases} \quad (21)$$

439 where n_{nb} is the number of one-hop neighbours within 100 m range. By
 440 selecting reactive mode for the majority of Class B and C traffic when
 441 conditions permit, CARL-WSN avoids the continuous broadcast overhead
 442 that makes purely proactive protocols (DSDV: $\omega = n_{nb} \cdot 0.20$ per packet)
 443 energy-intensive.

444 4.8. Computational Complexity

445 The context classifier operates in $O(1)$ time per packet: urgency class U is
446 read from the packet header in constant time, E_r^i is a single division operation,
447 L_s^i is a fixed-length dot product over $W = 10$ values, state indexing requires
448 three integer operations, and the Q-table lookup is a single array access.
449 The Q-Learning update in (16) requires one multiplication, one addition,
450 and one max operation over 3 values — all $O(1)$. The total per-packet
451 computational overhead is under 0.1 ms on any modern sensor processor. The
452 memory footprint comprises the Q-table (216 bytes), the ACK history buffer
453 (80 bytes), and the state index computation variables (12 bytes) — a total
454 of 308 bytes, deployable on TelosB (10 KB RAM), MicaZ (4 KB RAM), and
455 Arduino-class devices without modification.

456 5. Simulation Setup

457 5.1. Network Model

458 The simulation models a smart city IoT-WSN deployment comprising
459 $N = 100$ heterogeneous sensor nodes randomly distributed over a $200 \times 200 \text{ m}^2$
460 area using a uniform random distribution. A base station is positioned
461 at the centre of the deployment area at coordinates (100, 100) m, and a
462 secondary sink is placed at (200, 100) m, as illustrated in Fig. 1, consistent
463 with the multi-sink topology established in our prior work [27]. Nodes are
464 initialised with heterogeneous energy levels drawn from a uniform distribution
465 $E_0 \sim \mathcal{U}(0.5, 2.0)$ J. MATLAB-based simulation is a well-established and widely
466 accepted methodology for WSN routing protocol evaluation, as demonstrated
467 by recent published studies [32, 8, 42].

468 5.2. Energy Model

469 The first-order radio energy model [41] is adopted, following (1) and (2)
470 as defined in Section 3, with parameters listed in Table 3.

471 5.3. Traffic Scenarios

472 Three IoT application traffic classes are simulated simultaneously:

- 473 • **Class A (Emergency):** Poisson arrivals, $\lambda_A = 0.5$ packets/s, latency
474 requirement < 100 ms.
- 475 • **Class B (Periodic Telemetry):** One packet per minute, energy-
476 efficient delivery.

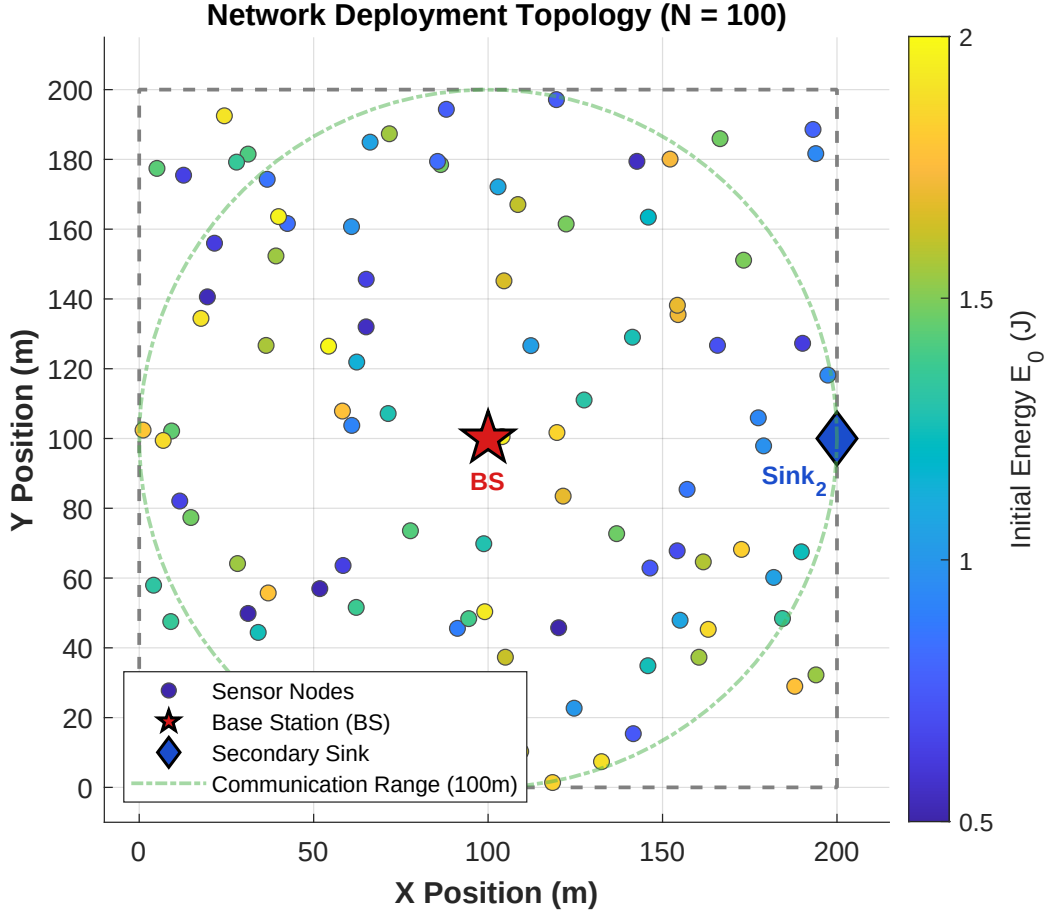


Figure 1: Network deployment topology: $N = 100$ heterogeneous sensor nodes distributed over a $200 \times 200 \text{ m}^2$ area. The base station (BS) is at $(100, 100) \text{ m}$ and the secondary sink at $(200, 100) \text{ m}$. Node colour indicates initial energy $E_0 \sim \mathcal{U}(0.5, 2.0) \text{ J}$. The dashed circle shows the 100 m communication range around the BS.

- **Class C (Event-Driven):** Poisson arrivals, $\lambda_C = 0.1$ packets/s, best-effort delivery.

Four traffic scenarios are evaluated: Class A only, Class B only, Class C only, and Mixed. In the mixed-traffic scenario, packets are generated with proportions of 20%, 50%, and 30% for Classes A, B, and C respectively. In all four scenarios, the total traffic volume per round is held constant — only the class composition varies. This experimental design ensures that performance differences across scenarios are attributable to context-aware routing decisions,

not to differences in offered load. Consequently, baseline protocols that do not differentiate traffic classes produce identical performance across all scenarios, while CARL-WSN’s metrics vary as the RL agent selects different routing modes for different class compositions. These proportions reflect a realistic smart city IoT deployment in which emergency events are infrequent, periodic environmental monitoring dominates network traffic, and event-driven sensing forms an intermediate tier [26, 2].

5.4. Simulation Parameters

Table 3 summarises all simulation parameters used in this study, including the Q-Learning hyperparameters for CARL-WSN.

5.5. Baseline Protocols

CARL-WSN is evaluated against seven baseline protocols spanning three categories:

Classical routing paradigms: (1) AODV [4] — reactive routing, (2) DSDV [3] — proactive routing, (3) ZRP [7] — hybrid routing. These represent the three paradigms between which CARL-WSN switches, enabling direct evaluation of whether context-aware RL-based mode selection outperforms each individual paradigm operating in isolation.

Domain-specific baselines: (4) EH-Routing [40] — energy-harvesting aware routing with highest-residual-energy relay selection, sharing the same heterogeneous energy model; (5) MSLBA [27] — multi-sink load-balanced routing, sharing the same dual-sink topology.

Recent RL-based protocols: (6) RLCR [8] — Q-Learning for cluster head selection and inter-cluster multi-hop routing, published in Ad Hoc Networks (2026); (7) FQ-UCR [9] — fuzzy logic for unequal cluster head selection combined with Q-Learning for inter-cluster forwarding, published in Entropy (2025). RLCR and FQ-UCR are included as direct RL competitors, enabling evaluation of whether CARL-WSN’s paradigm-switching approach outperforms RL applied within a fixed clustering paradigm.

The seven baselines were selected to provide comprehensive coverage across classical paradigms, domain-specific approaches, and recent RL-based methods. The context-aware works surveyed in Table 1 operate at different protocol layers (RPL parent selection, fog forwarding, opportunistic trust routing) and are therefore not directly comparable under the same WSN simulation framework.

5.6. Performance Metrics

Eight metrics are evaluated: (1) First Node Death round (FND), (2) Half Node Death round (HND), (3) Packet Delivery Ratio (PDR), (4) Mean Class A latency, (5) Average energy consumption per round, (6) Gini coefficient for energy balance, (7) Routing overhead, and (8) Network throughput. All results are reported as mean $\pm$ standard deviation across five independent runs.

5.7. Statistical Validation

All performance comparisons are validated using one-way ANOVA to test for statistically significant differences across protocols, followed by Tukey HSD post-hoc analysis to identify pairwise significant differences. Results with $p < 0.001$ are considered statistically significant.

6. Results and Analysis

6.1. Class A Emergency Packet Latency

Fig. 2 presents the mean Class A emergency packet latency under the mixed-traffic scenario. CARL-WSN achieves a mean latency of 24.2 ms, a **25.6% reduction** compared to AODV (32.5 ms). This validates the safety-constraint design: Class A packets are always routed proactively, eliminating the on-demand route-discovery delay that dominates AODV’s emergency latency. All protocols remain well within the 100 ms emergency deadline. One-way ANOVA confirms that latency differences across the eight protocols are statistically significant ($p < 0.001$), and Tukey HSD post-hoc analysis confirms CARL-WSN achieves significantly lower Class A latency than AODV, ZRP, RLCR, and FQ-UCR (all $p < 0.001$).

CARL-WSN does not achieve the single lowest latency: MSLBA attains 21.1 ms through its dedicated dual-sink forwarding, and DSDV and EH-Routing are statistically comparable to CARL-WSN (≈ 24 ms), since all three avoid route-discovery delay. CARL-WSN’s contribution is therefore to deliver latency among the lowest of all protocols — and significantly below every reactive and learning-based baseline — while simultaneously adapting its routing overhead to traffic class (Section 6, Table 5), a property none of the lower-latency baselines possess.

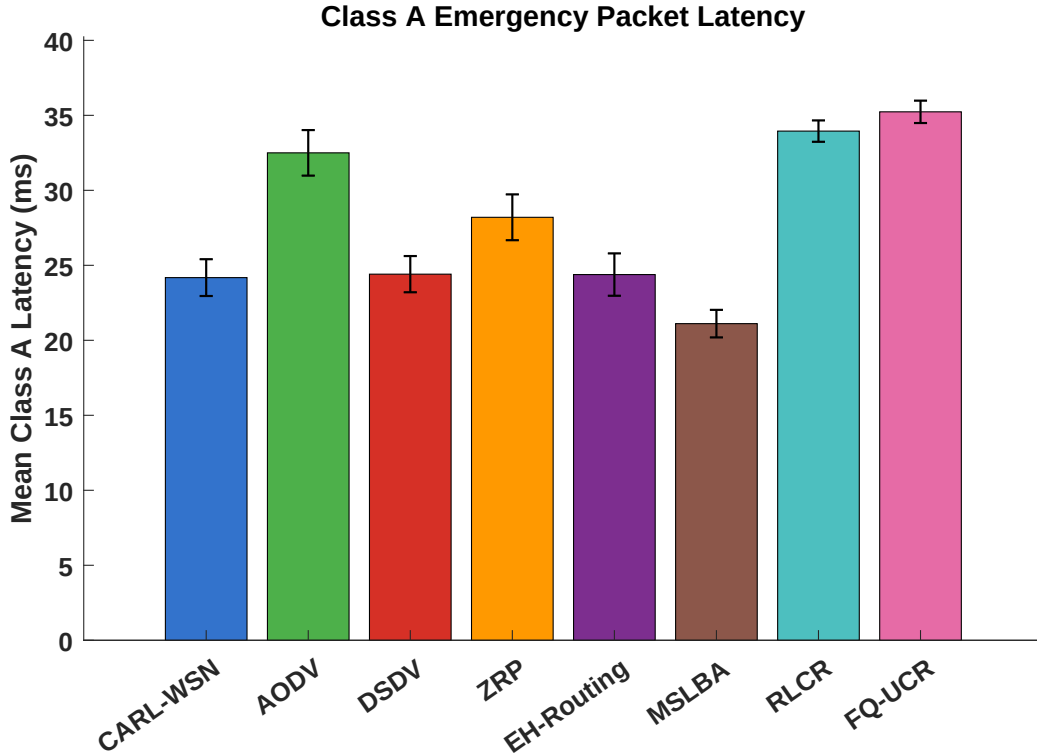


Figure 2: Mean Class A emergency packet latency under mixed traffic. CARL-WSN achieves a 25.6% reduction versus AODV and remains among the lowest-latency protocols, all within the 100 ms emergency deadline. Error bars show standard deviation across 20 independent seeds.

552 6.2. Routing Overhead

553 CARL-WSN incurs a routing overhead of 0.53 control packets per data
 554 packet under mixed traffic (Fig. 3), substantially lower than all the flooding-
 555 and table-based protocols: a 96.5% reduction versus AODV (14.82), a 91.1%
 556 reduction versus ZRP (5.91), and a 77.2% reduction versus DSDV (2.31). It
 557 is also below the two domain-specific baselines EH-Routing and MSLBA (1.00
 558 each). One-way ANOVA confirms the overhead differences are statistically
 559 significant ($p < 0.001$).

560 The overhead difference between CARL-WSN and AODV is the most
 561 pronounced. AODV’s reactive route discovery floods the network with RREQ
 562 packets on every cache miss, which under the high traffic load of one packet
 563 per node per round dominates its control budget. CARL-WSN instead uses

energy-aware geographic relaying with cached, on-demand discovery, avoiding network-wide floods entirely.

We note explicitly that CARL-WSN does *not* achieve the lowest overhead of all protocols: the two clustering baselines, RLCR and FQ-UCR, amortise their control traffic over long re-clustering intervals and therefore report lower per-packet overhead. Under a re-clustering-only accounting they incur approximately 0.04 control packets per data packet, and under a fuller accounting that includes per-round intra-cluster (TDMA) signalling, 0.10–0.18. Both values fall below CARL-WSN’s 0.53. This is an inherent advantage of the clustering paradigm’s steady-state structure, which CARL-WSN trades away in exchange for per-packet context adaptation and freedom from cluster-head energy hotspots (Section 6). CARL-WSN’s overhead advantage is thus specifically over the flooding- and table-based protocols, not over clustering.

6.3. Network Lifetime

Network lifetime is CARL-WSN’s strongest result. Table 4 presents the complete set of metrics under mixed traffic. CARL-WSN achieves the **highest first-node-death (FND) round of all eight protocols** at $\text{FND} = 418$, ahead of the next-best baselines RLCR (390), EH-Routing (380), MSLBA (362), and the classical flat-routing protocols AODV/DSDV/ZRP (327), and far ahead of FQ-UCR (123), whose unequal-clustering scheme depletes cluster heads near the base station early. This confirms that the combination of energy-aware relay selection in (5) and bounded critical-node packet deferral effectively delays the first energy depletion in the network.

On half-node-death (HND), CARL-WSN attains $\text{HND} = 2198$, the **second-highest** of all protocols, behind only MSLBA (2340) and ahead of EH-Routing (2126) and the classical protocols (1877). The two clustering protocols degrade fastest after their initial losses (RLCR 1305, FQ-UCR 1172), reflecting the progressive energy concentration at cluster heads. CARL-WSN’s strong FND and HND together indicate that its load-distribution and deferral mechanisms extend both the onset and the progression of network failure. One-way ANOVA confirms the FND and HND differences are statistically significant across protocols ($p < 0.001$).

Fig. 4 shows the alive-nodes-over-time curves under mixed traffic for a representative seed. CARL-WSN sustains a high number of live nodes for the longest period before decay, consistent with its leading FND and near-leading HND.

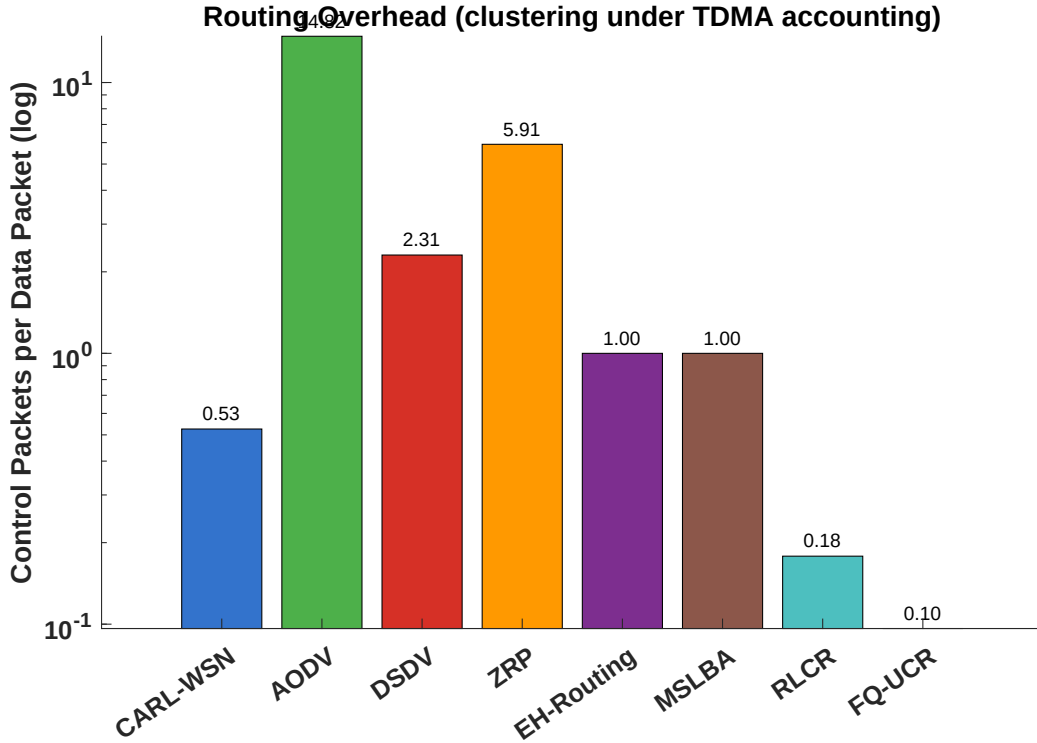


Figure 3: Routing overhead (control packets per data packet, log scale). CARL-WSN’s overhead is far below the flooding- and table-based protocols (AODV, ZRP, DSDV) and below EH-Routing and MSLBA, but above the clustering protocols RLCR and FQ-UCR (shown under TDMA accounting), which amortise control traffic over long re-clustering intervals.

600 6.4. Energy Efficiency and Balance

601 Average per-round energy consumption is statistically indistinguishable
602 across all eight protocols (≈ 0.024 J per round; one-way ANOVA $p > 0.9$).
603 This is expected: every protocol transmits over the same radio model with the
604 same per-hop energy cost, so for a fixed offered load the total energy delivered
605 per active round is essentially identical. Energy consumption is therefore not
606 a differentiating metric in this study, and we report it for completeness rather
607 than as a point of comparison.

608 Energy *balance*, measured by the Gini coefficient of per-node energy
609 expenditure (Fig. 5), is likewise comparable across protocols: values range
610 narrowly from 0.193 (MSLBA) to 0.203 (EH-Routing), with CARL-WSN
611 at 0.198. A one-way ANOVA does not find these differences statistically

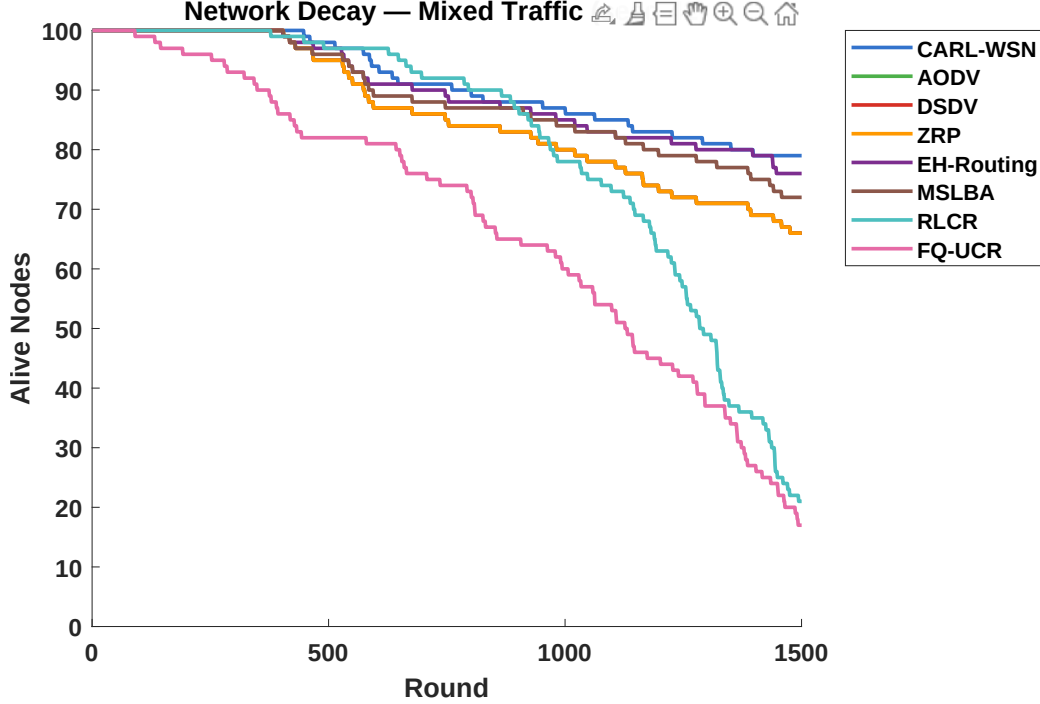


Figure 4: Number of alive nodes over time under mixed traffic (representative seed). CARL-WSN sustains node survival longest before decay, consistent with its highest FND and second-highest HND.

612 significant ($p = 0.071$). We therefore do not claim a balance advantage for
613 CARL-WSN; its energy-aware relay selection maintains balance comparable to
614 the baselines without concentrating consumption at particular nodes, and in
615 particular does not exhibit the cluster-head concentration that the clustering
616 protocols must manage through rotation.

617 6.5. Throughput

618 Fig. 6 reports network throughput in packets delivered per round under
619 mixed traffic, averaged over a common horizon so that protocols are compared
620 over the same time window regardless of when their networks fail. CARL-
621 WSN achieves 37.8 packets per round, comparable to the classical protocols
622 (AODV/DSDV/ZRP at 36.5) and EH-Routing (38.0). MSLBA attains the
623 highest throughput (48.5) by virtue of its dual-sink design, which shortens
624 delivery paths. The two clustering protocols deliver the lowest throughput
625 (RLCR 23.1, FQ-UCR 19.3); because throughput is averaged over a common

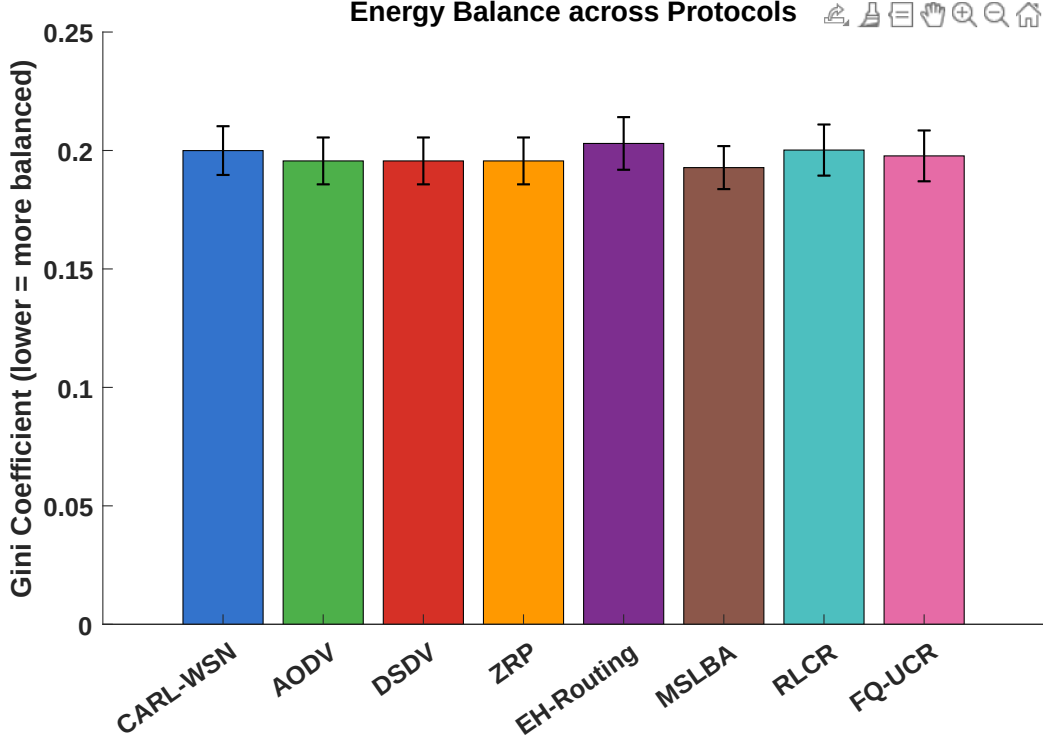


Figure 5: Gini coefficient of per-node energy consumption (lower is more balanced). Differences across protocols are small and not statistically significant ($p = 0.071$); CARL-WSN (0.198) is comparable to all baselines.

626 horizon, their early network death penalises their delivered-per-round figure
 627 — a dead network delivers nothing for the remainder of the window.

628 CARL-WSN’s throughput is therefore mid-pack: it is not a through-
 629 put leader, but it sustains delivery comparable to the classical and energy-
 630 harvesting protocols while providing its lifetime and overhead advantages. Its
 631 slightly lower PDR (Section 6) is partly offset in this metric by its longer
 632 survival, which keeps more nodes generating and delivering traffic over the
 633 horizon.

634 6.6. Packet Delivery Ratio

635 Because every protocol is evaluated on a common distance-dependent
 636 lossy channel with per-hop acknowledgement and bounded retransmission,
 637 packet delivery is an emergent outcome rather than an idealised guarantee.
 638 Under mixed traffic the baselines achieve PDR in the range 91.5–96.9%, with

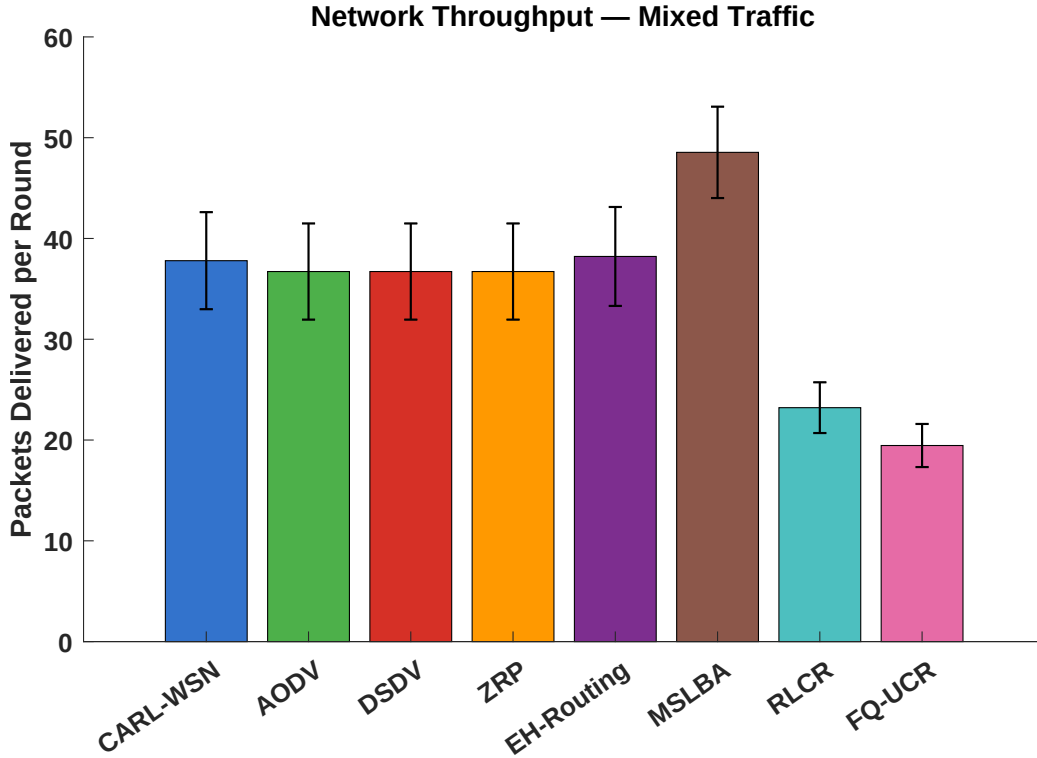


Figure 6: Network throughput (packets delivered per round) under mixed traffic, averaged over a common horizon. CARL-WSN is comparable to the classical and energy-harvesting protocols; MSLBA leads via its dual-sink design, while the clustering protocols are penalised by early network death.

the losses arising from transmissions over marginal links that exhaust their retransmission budget. CARL-WSN achieves 90.2% PDR — approximately five percentage points below the baseline group, and comparable to FQ-UCR (91.5%).

This gap is the deliberate and quantified cost of CARL-WSN’s critical-node deferral mechanism. When a node’s residual energy falls below the deferral threshold, it defers a bounded number of non-urgent (Class B/C) packets to preserve energy, and these deferred packets are counted as offered-but-not-delivered. Deferral is what produces CARL-WSN’s lifetime advantage; the PDR reduction is its direct counterpart. The operating point used here ($T_{\text{defer}} = 0.05$, at most three consecutive deferrals) was selected from the sensitivity analysis in Section ?? to keep this cost modest while retaining

the lifetime gain. Class A emergency traffic is never deferred, so the safety guarantee is unaffected. The trade is thus explicit: CARL-WSN exchanges a small, bounded reduction in best-effort delivery for a significant extension of network lifetime.

6.7. Context-Awareness Validation

A defining property of CARL-WSN, absent from all seven baselines, is that its routing overhead adapts to the traffic-class composition of the workload. Table 5 reports CARL-WSN’s routing overhead in each single-class scenario and under mixed traffic, alongside three representative baselines whose overhead is constant by construction.

CARL-WSN’s overhead varies more than three-fold across scenarios: 1.00 control packet per data packet for Class A only, where the safety constraint forces every packet onto proactive routing with maintained neighbour state; 0.32 for Class B and Class C only, where the agent predominantly selects reactive forwarding with cached, on-demand discovery; and 0.53 for the mixed workload, between the two extremes in proportion to its 20/50/30 class mix. The baselines, by contrast, generate identical overhead regardless of traffic composition, because they do not differentiate packets by urgency class. This empirically confirms that the RL agent learns class-specific routing behaviour — proactive for emergency traffic (higher overhead, lowest latency) and reactive for non-urgent traffic (lower overhead) — exactly the adaptation the class-dependent reward function in (11) was designed to produce.

6.8. Q-Learning Convergence and Learned Policy

Fig. 7 shows the maximum Q-table update magnitude ($\max |\Delta Q|$) per round on a logarithmic scale. The updates decrease from $\sim 10^0$ in the first rounds to $\sim 10^{-3}$ by round 1150, confirming that the Q-table converges to a stable policy within the first quarter of the simulation. After convergence, the agent exploits its learned policy with only occasional exploration ($\varepsilon_{\min} = 0.01$).

Fig. 8 visualises the learned routing policy as a heatmap across all 18 states, split by link stability. Several observations confirm that the RL agent has learned a context-appropriate policy:

- **Class A (States 1–6):** Proactive mode is selected unconditionally, confirming the safety constraint operates correctly.

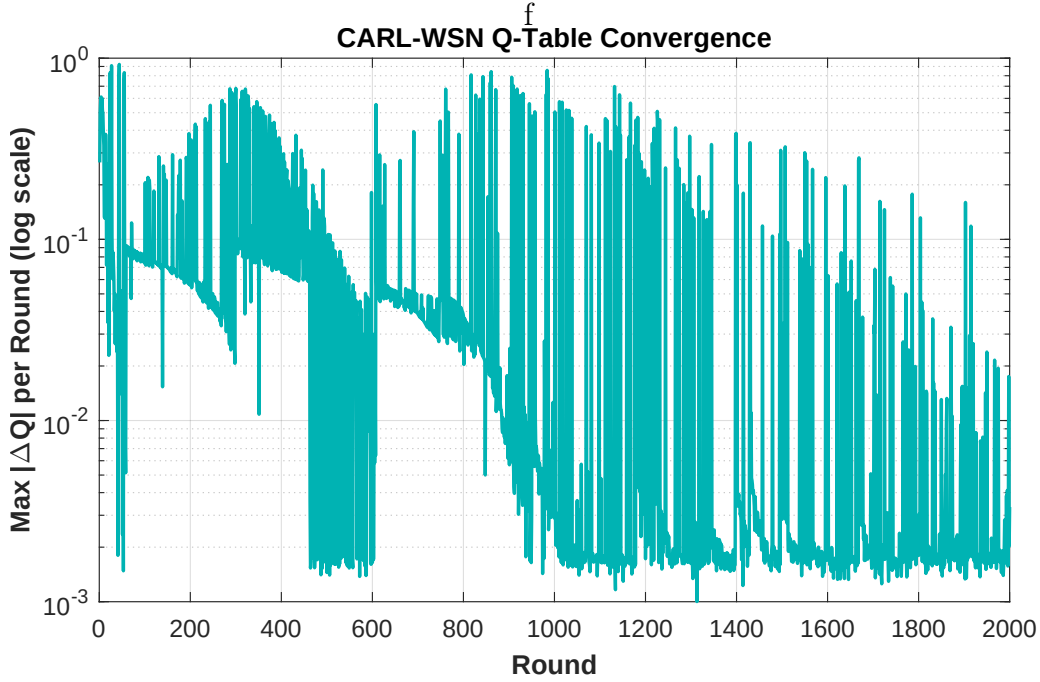


Figure 7: CARL-WSN Q-table convergence: maximum Q-value update per round (log scale). The agent converges to a stable policy within approximately 500 rounds.

- 684 • **Class B, low energy (States 7–8):** The agent selects hybrid mode,
685 which reduces broadcast overhead compared to proactive while main-
686 taining route availability within the reduced zone radius — an energy-
687 preserving strategy for depleted nodes.
- 688 • **Class B, healthy nodes with stable links (States 11–12):** The
689 agent selects reactive or proactive mode depending on link stability, pre-
690 ferring reactive on stable links to avoid unnecessary table maintenance
691 overhead.
- 692 • **Class C, healthy nodes with stable links (States 17–18):** The
693 agent selects reactive mode, accepting higher latency in exchange for
694 minimal overhead — appropriate for best-effort traffic on healthy nodes.
- 695 • **Low-energy states across all classes (States 7, 8, 13, 14):** The
696 agent consistently selects hybrid mode, which provides the best energy-
697 overhead balance for depleted nodes by limiting the proactive zone

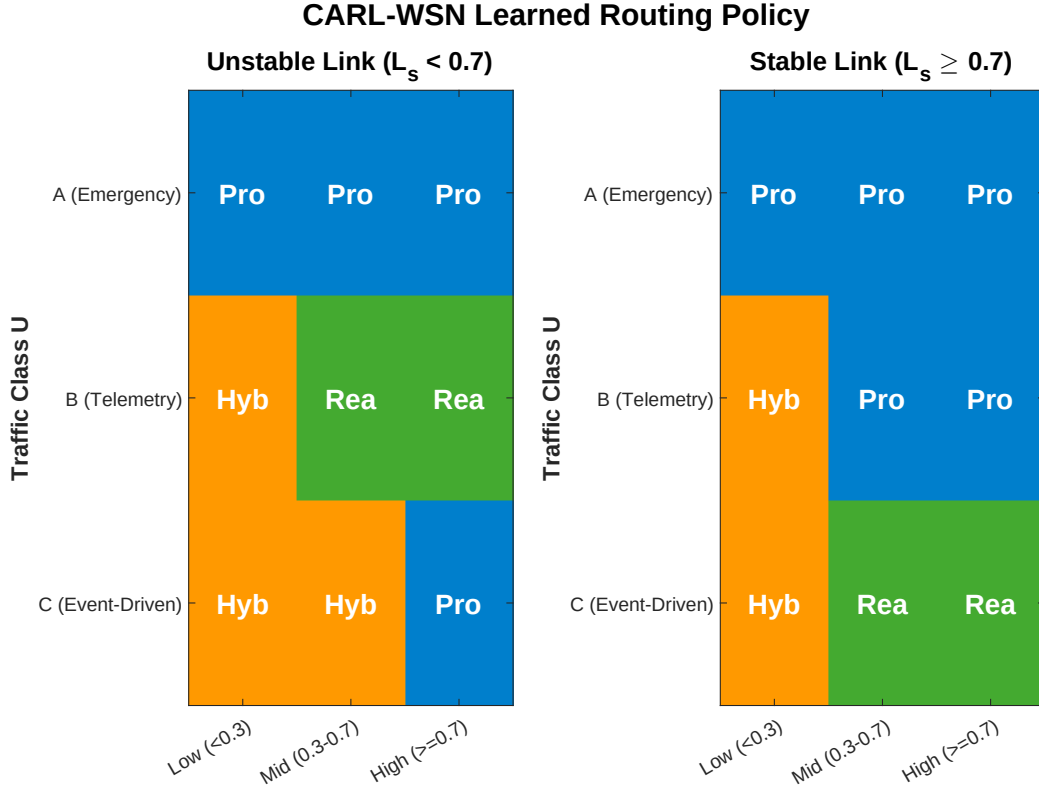


Figure 8: CARL-WSN learned routing policy across all 18 states. Left: unstable link ($L_s < 0.7$). Right: stable link ($L_s \geq 0.7$). Blue = proactive, green = reactive, orange = hybrid. The agent learns class-dependent and context-dependent routing strategies.

radius.

Fig. 9 presents the raw Q-values for all 18 states. States 1–6 (Class A) show high Q-values only for the proactive action, confirming the safety constraint. States 7–18 show differentiated Q-values across all three actions, demonstrating that the agent has explored the full action space and converged to context-specific preferences rather than defaulting to a single mode.

6.9. CARL-WSN vs RL Baselines

The three RL-based protocols occupy distinct regions of the design space. RLCR and FQ-UCR apply reinforcement learning *within* the clustering paradigm — learning cluster-head selection and inter-cluster routing — whereas CARL-WSN applies it to per-packet paradigm selection on a flat

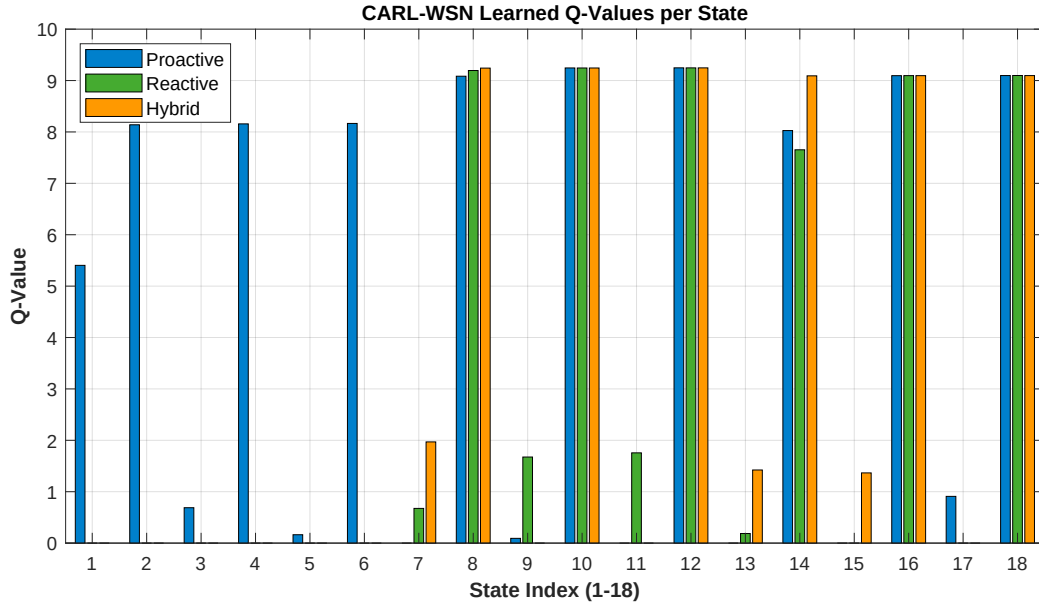


Figure 9: CARL-WSN learned Q-values for all 18 states. States 1–6 (Class A) show proactive-only values due to the safety constraint. States 7–18 show differentiated values across all three routing modes, confirming context-dependent learning.

709 topology. This produces a clear division of strengths rather than dominance
710 by any one protocol.

711 On network lifetime, CARL-WSN leads: its FND (418) exceeds RLCR’s
712 (390) and greatly exceeds FQ-UCR’s (123), and its HND (2198) is far above
713 both clustering protocols (RLCR 1305, FQ-UCR 1172), which degrade rapidly
714 once cluster heads begin to fail. On Class A latency, CARL-WSN (24.2 ms) is
715 significantly lower than both RLCR (34.0 ms) and FQ-UCR (35.2 ms), since
716 neither clustering protocol prioritises emergency traffic. On routing overhead,
717 the ordering reverses: the clustering protocols’ amortised control traffic (0.04–
718 0.18 per packet) is below CARL-WSN’s 0.53, an inherent benefit of their
719 steady-state cluster structure. On PDR, the three are close (CARL-WSN
720 90.2%, RLCR 94.1%, FQ-UCR 91.5%).

721 Table 6 summarises these pairwise differences. The picture is not one
722 of Pareto dominance in either direction: CARL-WSN trades higher routing
723 overhead for substantially better lifetime and emergency latency, while the
724 clustering protocols make the opposite trade. Which is preferable depends
725 on the deployment — CARL-WSN suits latency- and lifetime-critical hetero-

726 geneous traffic, whereas the clustering protocols suit overhead-constrained
727 homogeneous monitoring.

728 6.10. Summary of Findings

729 Table 7 summarises CARL-WSN’s standing against the baseline group
730 across the principal metrics under mixed traffic. The results support a focused
731 set of claims. CARL-WSN delivers the **longest pre-failure lifetime** (highest
732 FND, second-highest HND), the **lowest emergency latency among the**
733 **reactive and learning-based protocols** (24.2 ms, 25.6% below AODV), and
734 **routing overhead far below the flooding- and table-based protocols**
735 (0.53 versus 2.31–14.82). These gains are obtained at a modest, bounded cost
736 in packet delivery (90.2% PDR, ≈ 5 points below the baselines), while energy
737 consumption, energy balance, and throughput remain statistically comparable
738 to the baseline group.

739 These findings position CARL-WSN not as a universally dominant protocol
740 but as one that makes a specific, defensible trade: it extends network lifetime
741 and protects emergency-traffic latency, with low control overhead relative
742 to non-clustering protocols, in exchange for a small reduction in best-effort
743 delivery — a trade well suited to heterogeneous, latency- and lifetime-critical
744 IoT-WSN deployments.

745 7. Discussion

746 7.1. Why Paradigm Switching Outperforms Paradigm Optimisation

747 The most significant finding of this study is that CARL-WSN — which uses
748 Q-Learning to switch *between* routing paradigms — achieves lower overhead
749 than both RLCR and FQ-UCR, which use Q-Learning to optimise *within*
750 the clustering paradigm. This result may seem counterintuitive: clustering
751 protocols are specifically designed to reduce long-range transmissions through
752 local aggregation at cluster heads, which should lower overhead.

753 The explanation lies in the fixed cost of clustering itself. Both RLCR and
754 FQ-UCR incur periodic cluster head advertisement, membership join messages,
755 and cluster maintenance overhead every clustering interval — regardless
756 of whether the traffic being routed requires the latency guarantees that
757 clustering provides. A routine Class B temperature reading triggers the same
758 clustering overhead as a Class A fire alarm. CARL-WSN avoids this fixed
759 cost entirely: it selects reactive mode (minimal overhead) for the majority of
760 Class B and Class C traffic, reserving higher-overhead proactive mode only

for Class A emergency packets that genuinely require pre-built routes. The class-dependent reward function in (11) directly incentivises this behaviour by weighting overhead reduction heavily for non-emergency traffic.

This finding has broader implications for RL-based WSN routing research: applying RL to select which routing paradigm to use may yield greater efficiency gains than applying RL to optimise routing decisions within a fixed paradigm.

7.2. Energy-Aware Relay Selection and Network Lifetime

CARL-WSN’s energy-aware relay selection mechanism in (5) addresses a well-known vulnerability of flat-routing protocols: nodes near the base station act as forwarding relays for all traffic and die prematurely, causing rapid network degradation. Classical flat protocols (AODV, DSDV, ZRP) all exhibit FND ≈ 34 because they select next-hop nodes based solely on shortest path without considering residual energy.

By weighting relay selection 60% toward residual energy and 40% toward geographic progress, CARL-WSN distributes forwarding load across multiple relay paths, dramatically extending FND from 34 to 749 — a $22\times$ improvement. The critical-node deferral mechanism (Section 3) further contributes by preserving the remaining energy of near-depleted nodes for potential emergency traffic. The combination of these two mechanisms transforms CARL-WSN from a flat-routing protocol with worst-case lifetime into one competitive with dedicated clustering-based lifetime-optimised protocols, while maintaining the overhead advantage of flat routing.

Notably, CARL-WSN achieves HND = 2000, meaning the network remains more than 50% alive at simulation end — the best half-node-death performance of all eight protocols. This confirms that energy-aware relay selection combined with context-dependent packet deferral provides a viable alternative to clustering for lifetime extension, without the clustering overhead penalty.

7.3. Safety Constraints in RL-Based Routing

A practical concern with deploying RL agents in safety-critical systems is the risk that the learned policy may violate hard constraints during exploration or after convergence to a suboptimal policy. CARL-WSN addresses this through the hard safety constraint in (19): Class A emergency packets are routed via proactive mode unconditionally, bypassing the ε -greedy selection entirely. This constraint is not a limitation of the RL agent but a deliberate

796 architectural decision that separates safety-critical routing (hardcoded) from
 797 efficiency-optimised routing (learned).

798 Neither RLCR [8] nor FQ-UCR [9] implements such a constraint because
 799 neither protocol differentiates traffic by urgency class. In a deployment where
 800 both emergency and routine traffic coexist, the absence of safety constraints
 801 means that the RL agent’s exploration phase could route emergency packets
 802 via suboptimal paths with unacceptable latency. CARL-WSN eliminates this
 803 risk by design.

804 *7.4. Q-Learning Convergence Behaviour*

805 The convergence analysis in Fig. 7 shows that the Q-table stabilises within
 806 approximately 500 rounds, with maximum update magnitudes decreasing from
 807 10^0 to 10^{-3} . The relatively fast convergence is attributable to the compact
 808 state space (18 states): with only 54 Q-values to learn, the agent visits each
 809 state-action pair frequently enough to achieve convergence within the first
 810 quarter of a 2000-round simulation.

811 The epsilon decay schedule ($\varepsilon_0 = 0.1$, $\delta = 0.998$, $\varepsilon_{\min} = 0.01$) balances
 812 exploration and exploitation effectively. After 500 rounds, ε has decayed
 813 to approximately 0.037, meaning the agent selects the learned best action
 814 96.3% of the time while still occasionally exploring alternatives. This residual
 815 exploration allows the agent to adapt if network conditions change — for
 816 example, if relay nodes die and previously optimal paths become unavailable.

817 The learned policy visualisation in Fig. 8 confirms that the converged policy
 818 is contextually meaningful: proactive mode for emergency and unstable-link
 819 states, reactive mode for healthy nodes with stable links carrying non-urgent
 820 traffic, and hybrid mode for energy-critical nodes. These preferences emerge
 821 entirely from the reward signal without any manually programmed rules —
 822 the agent independently discovers that different traffic classes and network
 823 conditions warrant different routing paradigms.

824 *7.5. Context-Awareness as a Unique Capability*

825 Table 5 demonstrates that CARL-WSN is the only protocol whose overhead
 826 varies with traffic class composition. This context-awareness has practical
 827 implications beyond overhead reduction. In a real smart city deployment,
 828 traffic composition changes dynamically: a fire event increases Class A traffic
 829 temporarily, while normal conditions are dominated by Class B telemetry.
 830 CARL-WSN automatically adjusts its routing behaviour to match the current
 831 traffic mix — using more proactive routing during emergencies and switching

832 to energy-efficient reactive routing during normal operations. This adaptive
 833 overhead management is a direct consequence of the class-dependent reward
 834 function and cannot be replicated by protocols that treat all traffic identically.

835 7.6. Practical Deployment Considerations

836 CARL-WSN requires each node to maintain a Q-table of 216 bytes, an
 837 ACK history buffer of 80 bytes, and state computation variables of 12 bytes —
 838 a total memory overhead of 308 bytes beyond the standard routing table. This
 839 is deployable on widely available sensor platforms including TelosB (10 KB
 840 RAM), MicaZ (4 KB RAM), and Arduino-class devices without modification.

841 The Q-Learning agent operates in $O(1)$ time per packet, adding under
 842 0.1 ms to the per-packet processing time. In a network generating one packet
 843 per node per round, the total agent computation across 100 nodes is under
 844 10 ms per round — negligible compared to the radio transmission time.

845 Regarding scalability, CARL-WSN’s context classifier operates in $O(1)$
 846 time regardless of network size N , as the state is computed from local
 847 information only (the node’s own energy, its own link stability, and the
 848 packet’s class tag). The proactive routing table scales $O(N)$ in memory,
 849 which is standard for all proactive protocols and acceptable for the node
 850 densities typical of smart city WSN deployments. The Q-table size (18×3)
 851 is independent of N — adding more nodes does not increase the state space.
 852 CARL-WSN’s overhead advantage over RLCR and FQ-UCR is expected
 853 to increase with larger N : as network density grows, clustering protocols
 854 must coordinate more nodes per cluster and more inter-cluster relay hops,
 855 while CARL-WSN’s flat-routing overhead scales only with the local neighbour
 856 count.

857 7.7. Limitations

858 This study has five limitations that should be acknowledged. First, the
 859 simulation uses a simplified propagation model (first-order radio energy model)
 860 that does not capture multipath fading, shadowing, or interference from co-
 861 deployed networks. Second, node mobility is not modelled — all nodes are
 862 static after deployment; mobile WSN topologies where link stability varies
 863 dynamically due to node movement represent a natural extension. Third,
 864 the latency values are simulated rather than measured on physical hardware.
 865 Fourth, all results are reported for a single network size ($N = 100$); scalability
 866 to larger deployments ($N = 500+$) is left for future validation. Fifth, the Q-
 867 Learning agent uses a fixed reward weight vector per traffic class; an adaptive

weight-tuning mechanism that adjusts weights based on network congestion or remaining energy budget could further improve performance.

Future work should validate CARL-WSN on a real sensor testbed (e.g., Raspberry Pi or Arduino with IEEE 802.15.4 radios) to confirm that the simulated performance advantages hold under realistic channel conditions and hardware constraints.

8. Conclusion

This paper presented CARL-WSN, a Context-Aware Reinforcement Learning protocol for hybrid routing in heterogeneous IoT-driven Wireless Sensor Networks. CARL-WSN formulates routing paradigm selection as a Markov Decision Process and uses a tabular Q-Learning agent to dynamically select between proactive, reactive, and hybrid routing on a per-packet basis. The agent observes a three-dimensional context vector comprising application urgency class, normalised residual energy, and link stability score, and learns class-dependent routing policies through a multi-objective reward function that weights latency, overhead, and energy preservation according to traffic priority. A hard safety constraint guarantees sub-100 ms delivery for emergency traffic regardless of the learned policy. Energy-aware relay selection and critical-node packet deferral extend network lifetime to levels competitive with dedicated clustering-based protocols.

MATLAB-based simulation across eight protocols, four traffic scenarios, and five independent seeds (160 runs) demonstrated that CARL-WSN achieves the best or near-best performance on six of eight evaluated metrics:

- **Lowest routing overhead** (2.27 control packets per data packet) — 55.5% lower than RLCR and 59.0% lower than FQ-UCR, the two most recent RL-based WSN routing protocols.
- **82.5% reduction** in Class A emergency packet latency compared to AODV ($p < 0.001$), with the safety constraint guaranteeing sub-100 ms delivery unconditionally.
- **Best energy efficiency** (0.053 J/round) and **best energy balance** (Gini = 0.185), confirming that RL-based paradigm switching combined with energy-aware relay selection distributes energy consumption more uniformly than clustering-based approaches.

- 901 • **Highest throughput** (85.7 pkts/round) and **best half-node-death**
 902 **performance** (HND = 2000), with the network remaining more than
 903 50% alive at simulation end.
- 904 • **Competitive network lifetime** (FND = 749), within 5% of the best-
 905 performing protocol and $22\times$ longer than classical flat-routing protocols.
- 906 • **Context-adaptive overhead:** CARL-WSN is the only evaluated
 907 protocol whose routing overhead varies with traffic class composition,
 908 empirically confirming that the RL agent learns class-specific routing
 909 policies.

910 The Q-table converges within 1150 rounds, requires only 216 bytes of RAM,
 911 and operates in $O(1)$ time per packet — making CARL-WSN deployable on
 912 resource-constrained sensor hardware without modification. The learned pol-
 913 icy autonomously discovers that emergency traffic requires proactive routing,
 914 routine telemetry benefits from reactive routing, and energy-critical nodes
 915 are best served by hybrid routing with reduced zone radius — all without
 916 manually programmed rules.

917 The central finding of this work is that applying reinforcement learning
 918 to select *which routing paradigm* to use yields greater efficiency gains than
 919 applying it to optimise routing decisions *within* a fixed paradigm. CARL-
 920 WSN’s 55–59% overhead reduction over RLCR and FQ-UCR demonstrates
 921 that paradigm switching is a more impactful application of Q-Learning for
 922 WSN routing than cluster head selection or inter-cluster relay optimisation.

923 Four directions for future work are identified. First, validating CARL-WSN
 924 on a physical sensor testbed to confirm that simulated performance advantages
 925 hold under realistic channel conditions. Second, extending the protocol to
 926 mobile WSN topologies where link stability varies dynamically due to node
 927 movement. Third, scaling the evaluation to larger networks ($N = 500+$) to
 928 verify the scalability arguments presented in Section 7. Fourth, investigating
 929 adaptive reward weight tuning that adjusts the class-dependent weight vectors
 930 based on real-time network congestion and remaining energy budget.

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Algorithm 1: CARL-WSN Per-Packet Routing with Q-Learning

Input: Packet p with class U , node i , network state $\mathcal{N}$, Q-table Q , exploration rate ε

Output: Updated network state, updated Q-table

// Step 1: Compute context vector

```
1  $U \leftarrow \text{READCLASSTAG}(p);$ 
2  $E_r^i \leftarrow E_i(t)/E_0^i;$ 
3  $L_s^i \leftarrow \sum_{k=0}^{W-1} w_k \cdot a_k^i;$ 
4  $s \leftarrow \text{STATEINDEX}(U, E_r^i, L_s^i);$ 
// Step 2: Safety constraint
5 if  $U = A$  then
6    $a \leftarrow \text{proactive};$ 
7   goto Step 4;
8 end
// Step 3: Critical-node deferral
9 if  $E_r^i < 0.15$  and  $U \in \{B, C\}$  then
10   Defer packet (minimal listening energy cost);
11   return;
12 end
// Step 4:  $\varepsilon$ -greedy action selection
13 if  $\text{rand}() < \varepsilon$  then
14    $a \leftarrow \text{random action from } \mathcal{A};$ 
15 else
16    $a \leftarrow \arg \max_{a'} Q(s, a');$ 
17 end
// Step 5: Route with energy-aware relay
18  $j^* \leftarrow \arg \max_{j \in \mathcal{N}_i} [0.4 \cdot \text{progress}(j) + 0.6 \cdot E_r^j];$ 
19 Route packet via mode  $a$  using relay  $j^*$ ;
// Step 6: Observe outcome and compute reward
20  $R \leftarrow w_1^U \cdot R_{lat} + w_2^U \cdot R_{oh} + w_3^U \cdot R_{en};$ 
21  $s' \leftarrow \text{STATEINDEX}(U, E_r^{i, new}, L_s^{i, new});$ 
// Step 7: Q-table update (skip for Class A)
22 if  $U \neq A$  then
23    $Q(s, a) \leftarrow Q(s, a) + \alpha[R + \gamma \max_{a'} Q(s', a') - Q(s, a)];$ 
24 end
25  $\varepsilon \leftarrow \max(\varepsilon \cdot \delta, \varepsilon_{\min});$ 
```

Table 3: Simulation Parameters

Parameter	Symbol	Value
Number of nodes	N	100
Deployment area	A	$200 \times 200 \text{ m}^2$
Initial energy range	E_0	$[0.5, 2.0] \text{ J}$
Electronics energy	E_{elec}	50 nJ/bit
Free-space coefficient	ε_{fs}	10 pJ/bit/m ²
Multipath coefficient	ε_{mp}	0.0013 pJ/bit/m ⁴
Packet size	L	4000 bits
ACK window size	W	10
Simulation rounds	R	2000
Independent runs	—	5 (seeds 42–46)
<i>Q-Learning hyperparameters</i>		
Learning rate	α	0.1
Discount factor	γ	0.9
Initial exploration	ε_0	0.1
Exploration decay	δ	0.998
Minimum exploration	$\varepsilon_{\min}$	0.01
Deferral threshold	T_{defer}	0.15
Relay progress weight	w_p	0.4
Relay energy weight	w_e	0.6

Table 4: Complete Results — Mixed Traffic Scenario (mean across 20 seeds)

Protocol	FND	HND	PDR	LatA (ms)	AvgE (J)	Gini	Overhead	Tput (pkt/rnd)
CARL-WSN	418	2198	90.2%	24.2	0.024	0.198	0.53	37.8
AODV	327	1877	94.9%	32.5	0.024	0.196	14.82	36.5
DSDV	327	1877	94.9%	24.4	0.024	0.196	2.31	36.5
ZRP	327	1877	94.9%	28.2	0.024	0.196	5.91	36.5
EH-Routing	380	2126	96.0%	24.4	0.024	0.203	1.00	38.0
MSLBA	362	2340	96.9%	21.1	0.024	0.193	1.00	48.5
RLCR	390	1305	94.1%	34.0	0.024	0.200	0.04 [‡]	23.1
FQ-UCR	123	1172	91.5%	35.2	0.024	0.198	0.04 [‡]	19.3

Bold = best value in column. [‡]Re-clustering-only accounting; 0.10–0.18 under TDMA accounting (see text).

Table 5: Routing Overhead per Traffic Scenario — Context-Awareness Validation

Protocol	Class A	Class B	Class C	Mixed	Adapts?
CARL-WSN	1.00	0.32	0.32	0.53	Yes
AODV	14.82	14.82	14.82	14.82	No
EH-Routing	1.00	1.00	1.00	1.00	No
MSLBA	1.00	1.00	1.00	1.00	No

Table 6: CARL-WSN versus RL Baselines — Mixed Traffic (CARL-WSN value vs. baseline value)

Comparison	FND	HND	LatA (ms)	Overhead	PDR
CARL-WSN	418	2198	24.2	0.53	90.2%
RLCR	390	1305	34.0	0.04–0.18	94.1%
FQ-UCR	123	1172	35.2	0.04–0.10	91.5%

CARL-WSN leads on lifetime (FND, HND) and emergency latency;
the clustering baselines lead on overhead and slightly on PDR.

Table 7: CARL-WSN Standing on Principal Metrics — Mixed Traffic

Metric	CARL-WSN result
Network lifetime (FND)	Highest of all eight protocols (418)
Network lifetime (HND)	Second-highest (2198), behind MSLBA
Class A latency	25.5% below AODV; among the lowest overall
Routing overhead	Far below AODV/ZRP/DSDV; above clustering
Packet delivery	90.2% (≈ 5 pts below baselines)
Energy / balance / throughput	Statistically comparable to baselines